

CS & IT ENGINEERING

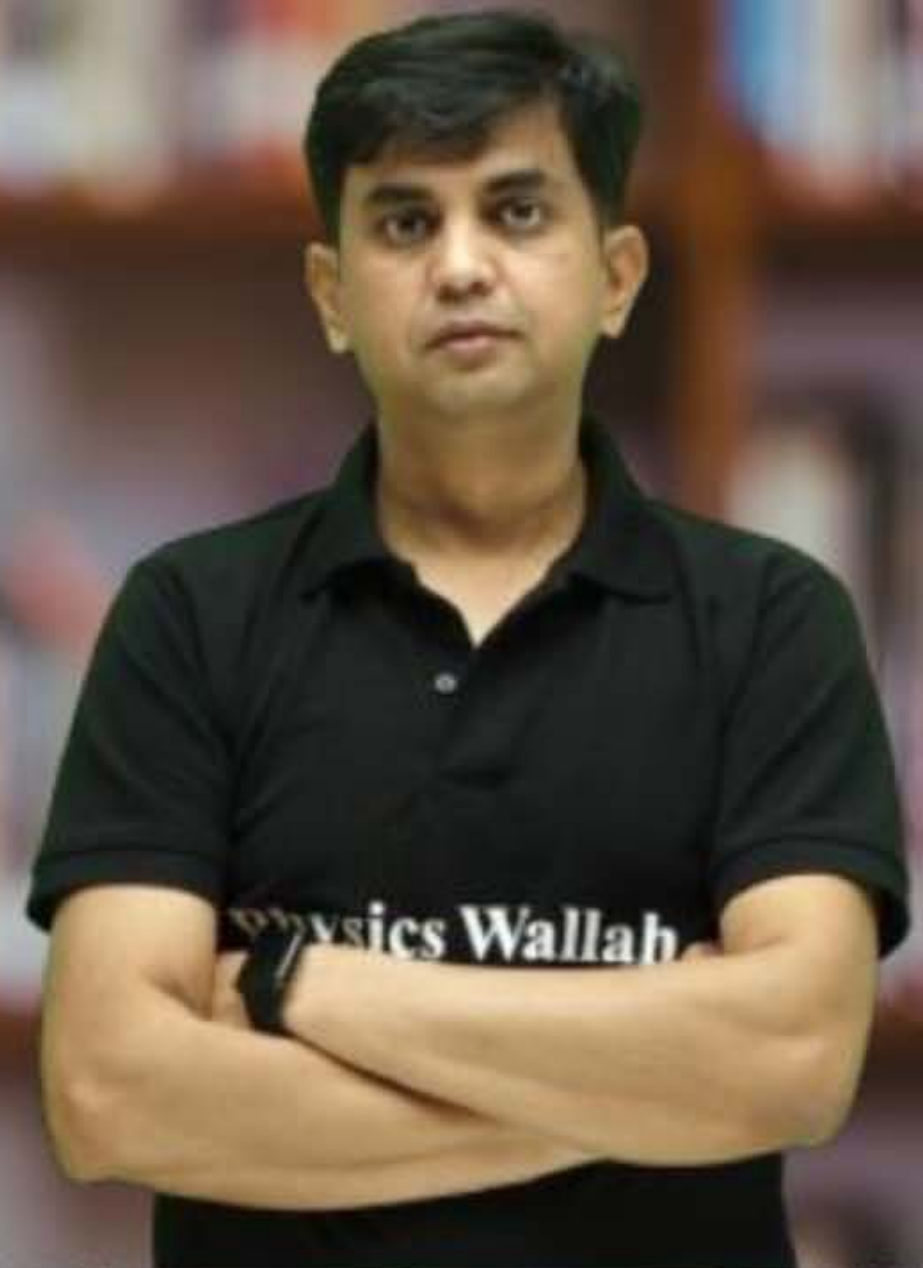


Computer Network

IPv4 Header

Lecture No. - 08

By - Abhishek Sir





Recap of Previous Lecture



Topic

IPv4 Options





Topics to be Covered



Topic

ICMP

Topic

DHCP

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
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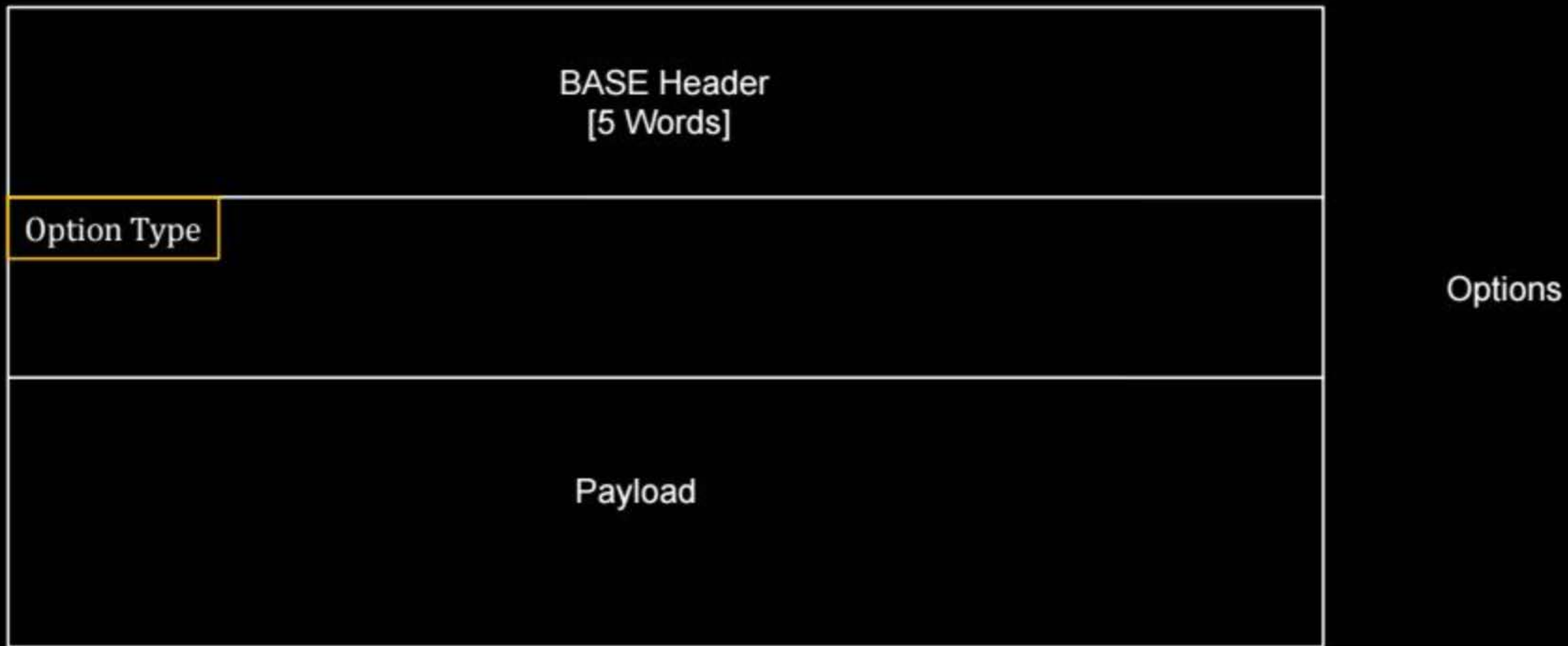
Topic : IPv4 Packet Header



0				16				31
VER	HLEN	Type of Services	Total Length					
Identification Number			0	D F	M F	Fragmentation Offset		
Time-to-Live		Protocol Type		Header Checksum				
Source IPv4 Address (32 - bits)								
Destination IPv4 Address (32 - bits)								
[Optional Header (Options)]								
Payload								



Topic : IPv4 Optional Header





Topic : Option Type



Number	Option Type
0	End of Option [Padding]
1	[No Operation]
7	Record Route
68	Timestamp
82	Traceroute
131	Loose Source Route
137	Strict Source Route



Topic : Source Routing



- Predetermine a route for the datagram
- Routing Path is determined by either "Source Host" or "Centralized Server"
- Routing Path is append in IPv4 options field by Source Host
- Source routing are of two types :
 1. Strict Source Routing
 2. Loose Source Routing



Topic : Strict Source Routing

- Complete path from [Source Router to Destination Router] [Sequence of IP address of intermediate routers]
- Datagram is required to follow exact defined route
- All of the routers defined must be visited by the datagram
- if the datagram visits a router that is not on the list then it discard the datagram and ICMP error message is generated



Topic : Strict Source Routing



Type = 1 00000001	Type = 137 10001001	Total Length = 39 00100111	Pointer = 4 00000100
First IP Address (Filled when started) (Source Router)			
Second IP Address (Filled when started)			
Third IP Address (Filled when started)			
Max ^m 9 Last IP Address (Filled when started) (Dest. Router)			

IPv4
Options



Topic : Loose Source Routing

- Similar to the strict source routing, but more relaxed
- All of the routers defined must be visited by the datagram
[in same order]
- In addition, the datagram can visit other routers as well



Topic : ICMP



→ Internet Control Message Protocol (ICMP)

→ To communicate network level information $\Rightarrow$ [Control Packet]
[Used by Hosts and Routers]

1. Error Reporting

[like Unreachable Network / Host / Protocol / Port]

2. Echo Request / Reply

[Used by "Ping"]



Topic : ICMP



→ Network Layer Protocol

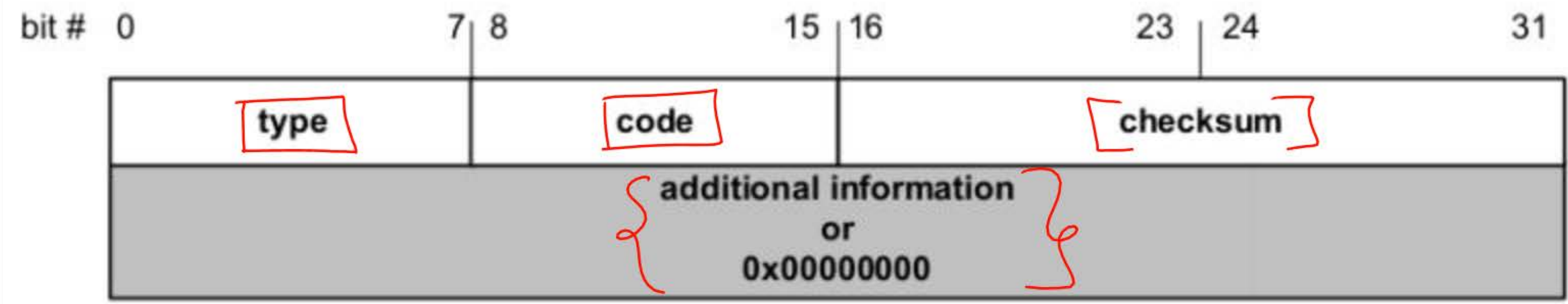
→ ICMP Messages carried in payload field of IP datagram

→ ICMP Message contains :

1. Type (1 byte)
2. Code (1 byte)



Topic : ICMP Message Format





Topic : ICMP Message Format



- Type (1 byte) : Type of ICMP message
- Code (1 byte) : Subtype of ICMP message
- Checksum (2 byte) : Similar to IP header checksum
[Checksum is calculated over entire ICMP message]
- Each ICMP messages is at least 8 bytes long
[If there is no additional data, 4 bytes set to zero]



Topic : ICMP Message Format



Type	Description
0	Echo Reply ✓
8	Echo Request ✓
→ 3	Destination Unreachable
11	Time Exceeded
30	Traceroute
4	Source Quench



Topic : Ping



- Application layer utility
- Used to test the reachability of a host on an IP network
- Ping also measures the round-trip time (RTT)
[messages sent from the originating host to a destination that are echoed back to the source]

Source Host

Ping

①

⑩

ICMP

②

⑨

IP

Destination
Device/Host/server



Ping

⑥

⑤

ICMP

⑦

④

IP

(ICMP Echo Reply)
⑧

③ (ICMP Echo Req)



Topic : ICMP Error Message



- ICMP error messages report error conditions
- Whenever IP datagram is discarded at Destination Host or Intermediate Router it sent "ICMP Error Message" to Source Host
- ICMP error messages contains complete IP header and the first 8 bytes of the IP datagram payload



Topic : Destination Unreachable

(Type = 3)

[Code]	Description
0	Network Unreachable
1	Host Unreachable
2	[Protocol Unreachable]
3	[Port Unreachable]
4	[Fragmentation Required] [DF Flag "Set"]

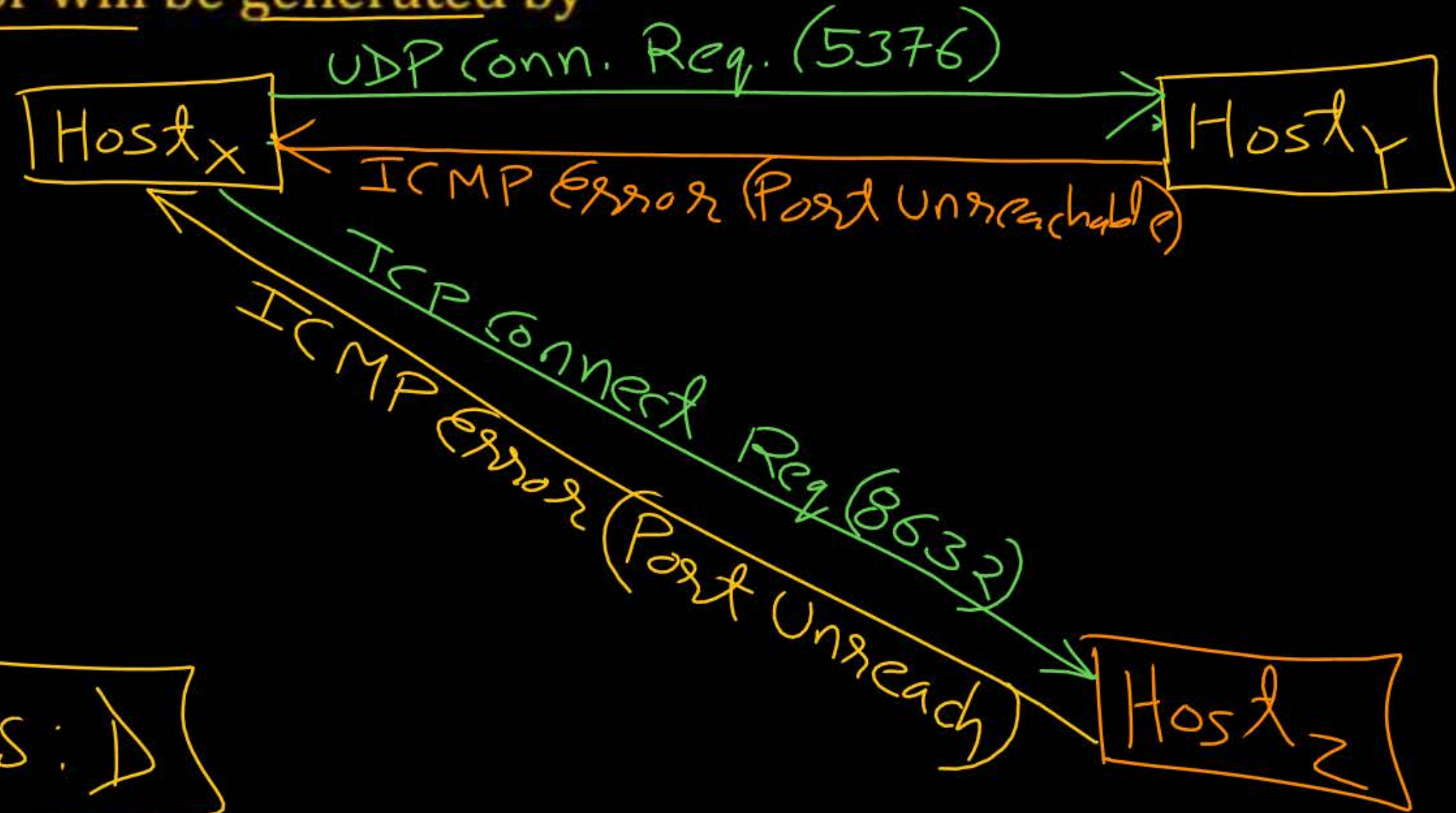
[MCQ]

[GATE-IT-2006] [2 Mark]



#Q. A program on machine X attempts to open a UDP connection to port 5376 on a machine Y, and a TCP connection to port 8632 on machine Z. However, there are no applications listening at the corresponding ports on Y and Z. An ICMP Port Unreachable error will be generated by

- A Y but not Z
- B Z but not Y
- C Neither Y nor Z
- ☒ D Both Y and Z



Ans: D



Topic : Traceroute

/ *tracert*



- Command-line utility
- Used to trace the route of network packets
[from source host to destination host]
- Identifies the IP addresses of routers along the path
- Valuable tool for troubleshooting network connectivity issues

[MCQ]

[GATE-IT-2005][1 Mark]



#Q. Trace-route reports a possible route that is taken by packets moving from some host A to some other host B. which of the following options represents the technique used by traceroute to identify these hosts?

- ☒ A By progressively querying routers about the next router on the path to B using ICMP packets, starting with the first router.
- ☐ B By requiring each router to append the address to the ICMP packet as it is forwarded to B. The list of all routers en-route to B is returned by B in an ICMP reply packet.
- ☐ C By ensuring that an ICMP reply packet is returned to A by each router en-route to B, in the ascending order of their hop distance from A
- ☐ D By locally computing the shortest path from A to B

Ans: A



Topic : DHCP ✓



- DHCP : Dynamic Host Configuration Protocol
- Dynamic assignment of IP address
[IP address assignment on demand]
- Avoid manual IP configuration
- Support mobility of laptops
- Earlier Protocols : RARP [Reverse ARP], BOOTP [Bootstrap Protocol]



Topic : DHCP



- Host dynamically obtains IP address from network server (DHCP Server)
[when it join networks]
- Can renew its lease on address in use
- Client - Server Model



Topic : DHCP Client – Server Scenario

- Typically DHCP server will be co-located in router
[serving all subnets to which routers is attached]
- DHCP runs over UDP
[Client uses port 68 and server uses port 67]
- Sometimes a subnet may have multiple DHCP servers

DHCP: App. layer Protocol

DHCP → UDP → IP



Topic : DHCP Operation



→ DHCP protocol is four step process.

1. DHCP Server Discovery
2. DHCP Server Offer(s)
3. DHCP Request
4. DHCP ACK



Topic : DHCP Server Discovery

- Newly arriving host (client) is to find DHCP server
- Client broadcast "DHCP discover" message

[Source IP Address]	: [0 . 0 . 0 . 0] : 68
[Destination IP Address]	: [255 . 255 . 255 . 255] : 67
yourIP	: 0 . 0 . 0 . 0
Type	: <u>DHCPDISCOVER</u>
<u>transaction ID</u>	: <u>752</u> → choose Randomly

[0 . 0 . 0 . 0]	: This Host	Limited
[255 . 255 . 255 . 255]	: <u>Network</u> / <u>Sub-network</u> Broadcast Address	



Topic : DHCP Server Offer



- When DHCP server receives "DHCP discover" message
- DHCP server responds by sending "DHCP offer" message

[Source IP Address]	: [210 . 210 . 210 . 2] : 67
[Destination IP Address]	: [255 . 255 . 255 . 255] : 68
<u>yourIP</u>	: [210 . 210 . 210 . 4]
Type	: <u>DHCPOFFER</u>
transaction ID	: 752
<u>DHCP Server IP</u>	: [210 . 210 . 210 . 2]
<u>Lifetime</u>	: 3600 sec



Topic : DHCP Request



- Newly arriving host (**client**) will choose from among one or more **server offers**
- **DHCP Client** responds to its selected offer with a "**DHCP request**" message

Source IP Address	: 0 . 0 . 0 . 0	: 68
Destination IP Address	: 255 . 255 . 255 . 255	: 67
<u>yourIP</u>	: 210 . 210 . 210 . 4	
<u>Type</u>	: DHCPREQUEST	
transaction ID	: 753	
DHCP Server IP	: 210 . 210 . 210 . 2	
<u>Lifetime</u>	: 3600 sec	

752 + 1



Topic : DHCP Acknowledgment

→ DHCP server responds to DHCP client with a “DHCP ACK” message

Source IP Address	: 210 . 210 . 210 . 210	: 67
Destination IP Address	: 255 . 255 . 255 . 255	: 68
<u>yourIP</u>	: 210 . 210 . 210 . 4	
Type	: DHCPACK	
transaction ID	: 753	
DHCP Server IP	: 210 . 210 . 210 . 2	
Lifetime	: 3600 sec	



Topic : DHCP Operation



=> DHCP server can return more than just allocated IP address on subnet

- Address of first hop router for client
- Name and IP address of DNS server
- Network Mask

=> DHCP server discovery and DHCP server offer(s) are optional steps
[if a client remember and wishes to reuse previously allocated network address]



2 mins Summary



Topic

ICMP

Topic

DHCP



THANK - YOU

